MIL-P-14990A(PA)
9 June 1975
SUPERSEDING
MIL-P-14990(MU)
4 February 1969

MILITARY SPECIFICATION

PLUGS, ELECTRICAL FOR LOCKING DEVICES XM77, XM85 and XM86

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of plug, entitled Plugs, Electrical for Locking Devices XM77, XM85 and XM86, which are used for training on the Nike Hercules warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8808362	- Packing and Marking for Storage and Shipment.
8880886	- Plug, Arming, Atomic Weapon, Training.
9210889	- Safe Plug Assembly, Electrical, Training.
ET-8880886	- Equipment Tabulation.
ET-9210888	- Equipment Tabulation.
ET-9210889	- Equipment Tabulation.

FSC 1115

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. *REQUIREMENTS*

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The plug shall comply with all requirements specified on the drawing (dwg) 8880886, 9210888 and 9210889, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Continuity and resistance. The resistance between connected pins listed in Table I of the plug shall be 0.003 ohm maximum (max) when being tested as specified in 4.4.1.

Table I

From Pin	To Pin	
(see dwg 888086)		
H	I	
K	L	
M	N	
N	O	
O	P	
(see dwg 9210888 and 9210889)		
A	M	
G	O	
E	F	
F	N	
I	J	Not applicable
J	K	to 9210889

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3.6 Dielectric strength. The dielectric strength of the circuits listed in Table II shall withstand a potential of 500 ± 50 volts root-mean-square (rms), 60 ± 5 Hertz (Hz) applied for 15 seconds (min) without breakdown or pitting of materials when tested as specified in 4.4.2.

Table II

From Pin	To Pin
(see dwg 8880886)	
All Pins	Shell
H	K,M
K	M
(see dwg 9210888 and 9210889)	
All Pins	Shell
I	F,G,A (Not applicable to 9210889)
F	G,A
G	A

3.7 Insulation resistance. The resistance between the pins listed in Table II shall be 100 megohms minimum (min) when being tested as specified in 4.4.3.

3.8 Soldering. All soldering shall be in compliance with MIL-STD-1460(MU).

3.9 Mating and unmating. The plug shall withstand 50 mating and unmating operations without evidence of external damage. After being manipulated as described in 4.4.4, the plug shall be tested for compliance with 3.5, 3.6 and 3.7.

3.10 Environmental testing (see 4.5).

3.10.1 Low temperature. The plug shall withstand a temperature of -65 ± 5 degrees Fahrenheit (°F) for 4 hours min and until the temperature of the plug has stabilized (see 6.4) without evidence of external damage. After being subjected as specified in 4.4.5.1, the plug shall be tested for compliance with the requirements of 3.5, 3.6 and 3.7.

3.10.2 Humidity. The plug shall withstand four 24 hours temperature-humidity cycles having a temperature range from 160 ± 5°F to 80 ± 20°F, and a relative humidity of 95 ± 5 percent without evidence of external damage. After being subjected as described in 4.4.5.2, the plug shall be tested for compliance with 3.5, 3.6 and 3.7.

3.10.3 Vibration. The plug shall withstand the environmental properties in Table III in each of its three mutually perpendicular axes without evidence of external damage. After being subjected as described in 4.4.5.3, the plug shall be tested for compliance with 3.5, 3.6 and 3.9.

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Table III

Vibration Test Schedule

Frequency (Hz)	Amplitude	Sweeptime + 1/2 Minute Cycle #1 80+20°F	Cycle #2 160+5°F	Cycle #3 -65+5°F
5-10-5	0.08 ± 0.008 Double Amplitude - inch	5	2.5	2.5
10-15-10	0.4 ± 0.04 Gravity unit peak	5	2.5	2.5
15-52-15	0.036 ± 0.0036 Double amplitude	5	2.5	2.5
52-500-52	0.5 ± 0.05 Gravity unit peak	5	2.5	2.5

3.10.4 Mechanical shock. The plug shall withstand a force of 50 ± 5 gravity units (g) peak amplitude, three times in each direction along its three mutually perpendicular axes (total 18 shocks) without evidence of external damage. Each shock shall last 11 ± 1 milliseconds (ms) as measured between peak points. The time to peak shall be reached within 5.5 ms. After being subjected as specified in 4.4.5.4, the plug shall be tested for compliance with 3.5 and 3.7.

3.10.5 High temperature. The plug shall withstand a temperature of 160 ± 5°F for 4 hours min and until the temperature of the plug has stabilized, without evidence of external damage. After being subjected as described in 4.4.5.5, the plug shall be tested for compliance with 3.5 and 3.7.

3.11 Workmanship.

3.11.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.11.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.11.3 Burrs. Parts shall be free from burrs.

3.11.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.11.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwgs 8880886, 9210888 and 9210889) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table IV. Tests shall be conducted in order specified in Table IV.

Table IV

First article inspection and tests

Inspection or Test		Dest	Non-Dest
1.	Classification of characteristics (see 4.3.2.1)		X
2.	Continuity and resistance (see 3.5)		X
3.	Dielectric strength (see 3.6)	X	
4.	Insulation resistance (see 3.7)		X
5.	Mating and unmating (see 3.9)	X	
6.	Vibration (see 3.10.3)	X	
7.	Low temperature (see 3.10.1)		X
8.	High temperature (see 3.10.5)		X
9.	Mechanical shock (see 3.10.4)	X	
10.	Humidity (see 3.10.2)	X	

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table V and Table VI), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table V
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

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Table VI
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Safe plug assembly, electrical (see dwgs 8880886, 9210888 and 9210889).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Failure to mate with connector 8801719	Gage
102.	Damaged finish exposing base metal	Visual
103.	Foreign matter on pins	Visual
104.	Threads damaged	Visual
105.	Plating missing from pins	Visual
106.	Insulating material cracked	Visual
107.	Soldering improper (see 3.8) prior to potting	Visual
*108.	Height of pins incorrect (pins of the same diameter must all appear to be within 1/16 inch in height relative to adjacent pin)	Gage
109.	Potting missing, improper or in threaded area	Visual
110.	Marking missing, improper or illegible	Visual
111.	Objective evidence relative to material missing or improper (see 3.1)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.11)	Visual

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4.3.2.1.2 Packing and marking for storage and shipments (see dwg 8808362) before closure and strapping.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Marking improper, illegible or missing for intermediate packaging (see dwg 8883539-4)	Visual
102.	Intermediate container improper	Visual
103.	Unit of product improperly protected	Visual
104.	Net load of intermediate container exceeded	Gage
Minor:		
201.	Evidence of poor workmanship (see 3.11)	Visual

4.3.2.1.3 Packing and marking for storage and shipment (see dwg 8808362).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Exterior shipping container size exceeded	Gage
102.	Gross weight of exterior container exceeded	Gage
103.	Container closure improper	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.11)	Visual

*In the event pin(s) do not appear to be within 1/16 inch in height relative to the adjacent pins, Gage (see 4.3.4) shall be used to determine the variation from the normal height. Reject any pin that varies by more than 0.063 inch max. from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface of all measurements of pin heights.

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4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Continuity and resistance	Major
b. Insulation resistance	Major
c. Dielectric strength	Critical
d. Mating and unmating	Critical
e. Low temperature	Major
f. Vibration	Major

Of these tests, c, d, and f are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Continuity and resistance, and insulation resistance. Each plug shall be tested in accordance with 4.4.1 and 4.4.3. If any item fails to comply with 3.5 and 3.7, the item shall be rejected and removed from the lot.

4.3.3.2 Dielectric strength, mating and unmating, low temperature, and vibration. A sample shall be selected at random in accordance with Table VII. The sample shall be tested in accordance with 4.4.2, 4.4.4, 4.4.5.1 and 4.4.5.3. If any item fails to comply with 3.6, 3.9, 3.10.1 and 3.10.3, the lot shall be rejected.

Table VII

Sample sizes for non-destructive testing

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). Els for assemblies are listed on the Equipment Tabulation, ET-8880886, 9210888 and 9210889. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part of Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Continuity and resistance. The circuit resistance shall be measured with an approved tester to determine compliance with 3.5.

4.4.2 Dielectric strength. The dielectric strength shall be measured with an approved tester while being tested to determine compliance with 3.6.

4.4.3 Insulation resistance. The insulation resistance shall be measured with an approved tester (megohm meter type) at a potential of 500 + 50 volts direct current (vdc) and current shall not exceed 5 milliamperes (ma). The potential shall be applied for 5 seconds minimum while testing to determine compliance with 3.7.

4.4.4 Mating and unmating. The plug shall be mated and unmated to the assembly illustrated in Figure 1, fifty times without the use of tools. After the 50th operations the plug shall be tested to determine compliance with 3.5, 3.6 and 3.7.

4.4.5 Environmental testing (see 3.10).

4.4.5.1 Low temperature. The plug shall be placed into an approved chamber and the temperature lowered to -65 + 5°F; maintained for 8 hours min and until the temperature of the plug has stabilized (see 6.4). After stabilization the plug shall be tested to determine compliance with 3.5 and 3.7.

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4.4.5.2 Humidity. The plug shall be put into an approved test chamber and subjected to the test requirements in 3.10.2. Each 24 hours humidity temperature cycle shall be regulated as follows:

(a) Uniformly raise the temperature from 80 ± 20°F to 160 ± 5°F (95 ± 5% RH) during first 2 hours.

(b) Maintain the temperature of 160 ± 5°F (95 ± 5% RH) for next six hours.

(c) Uniformly lower the temperature from 160 ± 5°F to 80 ± 20°F (95 ± 5% RH) during next 16 hours.

The plug shall be subjected to 4 such cycles. At the completion of the 4th cycle the plug shall be removed from the chamber and dried by applying filtered compressed air for 5 minutes or until dry. After the plug has dried, it shall be visually examined for external damage (see 6.5) and tested to determine compliance with 3.5 and 3.7.

4.4.5.3 Vibration. The plug shall be placed into a temperature chamber and conditioned as described in 4.4.5.1 or 4.4.5.5 as appropriate. The chamber shall be mounted on a fixture capable of being orientated to vibrate the plug on its three axes. While the temperature of the plug is maintained stabilized (see 6.4) it shall be subjected to the vibration schedule in 3.10.3. After being vibrated, the plug shall be tested to determine compliance with 3.5, 3.6, 3.9 and examined for external damage (see 6.5).

4.4.5.4 Mechanical shock. The plug shall be mounted on an approved fixture. The fixture shall be capable of being orientated to shock the plug in its three mutually perpendicular axes. The mounted plug then shall be subjected to the test requirements of 3.5. After being shocked, the plug shall be tested to determine compliance with 3.7, 3.5 and visually examined for external damage (see 6.5).

4.4.5.5 High temperature. The plug shall be placed into an approved chamber and the temperature raised to 160 + 5°F, maintained for 4 hours and until the temperature of the plug is stabilized (see 6.4). After stabilization the plug shall be tested to determine compliance with 3.5 and 3.7.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The plug shall be packaged in accordance with dwg 8808362 using the specified moisture barrier.

5.1.2 Level C. The plug shall be packaged in accordance with dwg 8808362 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The plug shall be packed in accordance with dwg 8808362.

5.3 Marking. Marking shall be as specified in accordance with dwg 8808362.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 External damage. A defect, such as obliterated markings, damaged finish exposing base metal, crack or bent pins that is detectable by visual examination.

6.6 Supersession data. This specification supersedes the requirements of Purchase Description MIL-P-14990(MU), dated 4 February 1969.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1115-A062

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MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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